

Lab 8

Q1

Practice 1

```
1/1 v + [ ] [ ] Tilix: japoriona@cis106: -
japoriona@cis106: ~
15804 with these requirements. We do not solicit donations in locations
15805 where we have not received written confirmation of compliance. To SEND
15806 DONATIONS or determine the status of compliance for any particular
15807 state visit www.gutenberg.org/donate.
15808
15809 While we cannot and do not solicit contributions from states where we
15810 have not met the solicitation requirements, we know of no prohibition
15811 against accepting unsolicited donations from donors in such states who
15812 approach us with offers to donate.
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15814 International donations are gratefully accepted, but we cannot make
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15816 outside the United States. U.S. laws alone swamp our small staff.
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15818 Please check the Project Gutenberg web pages for current donation
15819 methods and addresses. Donations are accepted in a number of other
15820 ways including checks, online payments and credit card donations. To
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15823 Section 5. General Information About Project Gutenberg™ electronic works
15824
15825 Professor Michael S. Hart was the originator of the Project
15826 Gutenberg™ concept of a library of electronic works that could be
15827 freely shared with anyone. For forty years, he produced and
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15829 volunteer support.
15830
15831 Project Gutenberg™ eBooks are often created from several printed
15832 editions, all of which are confirmed as not protected by copyright in
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15834 necessarily keep eBooks in compliance with any particular paper
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15837 Most people start at our website which has the main PG search
15838 facility: www.gutenberg.org.
15839
15840 This website includes information about Project Gutenberg™,
15841 including how to make donations to the Project Gutenberg literary
15842 Archive Foundation, how to help produce our new eBooks, and how to
15843 subscribe to our email newsletter to hear about new eBooks.
15844
15845
japoriona@cis106: ~$
```

Practice 2

```
1/1 v + [ ] [ ] Tilix: japoriona@cis106: -
japoriona@cis106: ~
Honey Graham Ohs calories = 120
Honey Nut Cheerios calories = 110
Honey-comb calories = 110
Just Right Crunchy Nuggets calories = 110
Just Right Fruit & Nut calories = 140
Kix calories = 110
Life calories = 100
Lucky Charms calories = 110
Maypo calories = 100
Muesli Raisins, Dates, & Almonds calories = 150
Muesli Raisins, Peaches, & Pecans calories = 150
Mueslix Crispy Blend calories = 160
Multi-Grain Cheerios calories = 100
NutHoney Crunch calories = 120
Nutri-Grain Almond-Raisin calories = 140
Nutri-grain Wheat calories = 90
Oatmeal Raisin Crisp calories = 130
Post Nut. Raisin Bran calories = 120
Product 19 calories = 100
Puffed Rice calories = 50
Puffed Wheat calories = 50
Quaker Oat Squares calories = 100
Quaker Oatmeal calories = 100
Raisin Bran calories = 120
Raisin Nut Bran calories = 100
Raisin Squares calories = 90
Rice Chex calories = 110
Rice Krispies calories = 110
Shredded Wheat calories = 80
Shredded Wheat 'n'Bran calories = 90
Shredded Wheat spoon size calories = 90
Snacks calories = 110
Special K calories = 110
Strawberry Fruit Wheats calories = 90
Total Corn Flakes calories = 110
Total Raisin Bran calories = 140
Total Whole Grain calories = 100
Triples calories = 110
Trix calories = 110
Wheat Chex calories = 100
Wheaties calories = 100
Wheaties Honey Gold calories = 110
japoriona@cis106: ~$
```

Practice 3

```
Tilix: japariona@cis106: ~/Documents/sample_files/Csv
1: japariona@cis106: ~/Documents/sample_files/Csv
Volkswagen Rabbit;29.8;4;89.00;62.00;1845.;15.3;80;Europe
Volkswagen Rabbit;41.5;4;98.00;76.00;2144.;14.7;80;Europe
Volkswagen Rabbit C (Diesel);44.3;4;90.00;48.00;2085.;21.7;80;Europe
Volkswagen Rabbit Custom;29.0;4;97.00;78.00;1940.;14.5;77;Europe
Volkswagen Rabbit Custom;31.9;4;89.00;71.00;1925.;14.0;79;Europe
Volkswagen Rabbit Custom Diesel;43.1;4;90.00;48.00;1985.;21.5;78;Europe
Volkswagen Rabbit l;36.0;4;105.0;74.00;1980.;15.3;82;Europe
Volkswagen Scirocco;31.5;4;89.00;71.00;1990.;14.9;78;Europe
Volkswagen Super Beetle 117;0;4;97.00;48.00;1978.;20.0;71;Europe
Volkswagen Super Beetle;26.0;4;97.00;46.00;1950.;21.0;73;Europe
Volkswagen Type 3;23.0;4;97.00;54.00;2254.;23.5;72;Europe
Volvo 144ea;19.0;4;121.0;112.0;2868.;15.5;73;Europe
Volvo 145e (sw);18.0;4;121.0;112.0;2933.;14.5;72;Europe
Volvo 244DL;22.0;4;121.0;98.00;2945.;14.5;75;Europe
Volvo 245;20.0;4;130.0;102.0;3150.;15.7;76;Europe
Volvo 264gl;17.0;6;163.0;125.0;3140.;13.6;78;Europe
japariona@cis106:~/Documents/sample_files/Csv$ sort -o sorted_cars.csv cars.csv
japariona@cis106:~/Documents/sample_files/Csv$ wc -l sorted_cars.csv
wc: sorted_cars.csv: No such file or directory
japariona@cis106:~/Documents/sample_files/Csv$ sort -o sorted_cars.csv cars.csv
japariona@cis106:~/Documents/sample_files/Csv$ wc -l sorted_cars.csv
408 sorted_cars.csv
japariona@cis106:~/Documents/sample_files/Csv$ wc -m sorted_cars.csv
22663 sorted_cars.csv
japariona@cis106:~/Documents/sample_files/Csv$
```

Challenge questions

Q1

```
Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ wc -l access.log
100 access.log
japariona@cis106:~$
```

Q2

```
Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ head -n 1 access.log
2025-01-02 15:16:38 Eren /download FAIL
japariona@cis106:~$
```

Q3

```
Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ tail -n 1 access.log
2025-06-22 14:50:54 Edward /404 FAIL
japariona@cis106:~$
```

Q4

```
Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ head -n 15 -v access.log
==> access.log <==
2025-01-02 15:16:38 Eren /download FAIL
2025-02-19 19:50:15 Asimov /download SUCCESS
2025-01-22 17:56:00 Jet /login DENIED
2025-11-18 18:55:39 Spike /upload SUCCESS
2025-07-06 23:02:30 Rei /login DENIED
2025-05-22 18:18:45 Honami /download FAIL
2025-05-02 01:46:24 Arisu /404 SUCCESS
2025-05-05 00:30:57 Edward /settings DENIED
2025-04-19 16:20:31 Touka /settings FAIL
2025-02-28 15:59:57 Shinji /reset FAIL
2025-12-25 21:14:04 Ken /admin SUCCESS
2025-01-23 22:41:06 Haruka /dashboard FAIL
2025-09-06 04:55:58 Ken /upload SUCCESS
2025-05-05 14:31:33 Jet /404 FAIL
2025-10-01 02:12:40 Rize /logout DENIED
japariona@cis106:~$
```

Q5

```
Tilix: japariona@cis106: ~
1: japariona@cis106: ~
japariona@cis106:~$ tail -n 15 -v access.log
==> access.log <==
2025-10-09 19:11:43 Walter /upload DENIED
2025-05-23 21:37:40 Arisu /settings FAIL
2025-09-06 10:17:57 Haruka /404 SUCCESS
2025-10-20 12:25:44 Arisu /dashboard SUCCESS
2025-12-01 15:13:12 Jet /admin SUCCESS
2025-03-16 06:28:44 Edward /404 DENIED
2025-02-12 19:40:38 Jet /reset SUCCESS
2025-11-01 04:30:52 Jet /reset DENIED
2025-01-15 22:59:18 Eren /settings DENIED
2025-04-08 14:17:14 Kei /404 DENIED
2025-07-02 06:18:13 Edward /login DENIED
2025-02-19 05:43:40 Faye /logout DENIED
2025-09-25 16:08:56 Spike /logout DENIED
2025-11-25 01:07:36 Walter /download DENIED
2025-06-22 14:50:54 Edward /404 FAIL
japariona@cis106:~$
```

Q6

```
1: japoriona@cis106: ~  
japoriona@cis106:~$ cut -d' ' -f3 access.log  
Eren  
Asimov  
Det  
Spike  
Rei  
Honami  
Arisu  
Edward  
Touka  
Shinji  
Ken  
Haruka  
Ken  
Det  
Rize  
Julia  
Rei  
Honami  
Annie  
Setas  
Edward  
Annie  
Faye  
Haruka  
Spike  
Rize  
Rize  
Annie  
Faye  
Eren  
Touka  
Ken  
Ken  
Ein  
Rize  
Shinji  
Julia  
Ken  
Ken  
Ein  
Faye  
Shinji  
Arisu  
Annie  
Asimov
```

Q7

```
1: japoriona@cis106: ~  
japoriona@cis106:~$ sort -k3,3 -o sorted_access_log_by_username.log access.log  
japoriona@cis106:~$ cat sorted_access_log_by_username.log  
2025-04-04 21:44:10 Annie /download FAIL  
2025-04-25 22:26:14 Annie /upload FAIL  
2025-05-13 05:05:05 Annie /admin DENIED  
2025-09-22 23:25:30 Annie /upload FAIL  
2025-12-10 08:28:44 Annie /settings FAIL  
2025-01-19 13:53:56 Arisu /download SUCCESS  
2025-05-02 01:46:24 Arisu /404 SUCCESS  
2025-05-18 13:00:10 Arisu /upload SUCCESS  
2025-05-23 21:37:40 Arisu /settings FAIL  
2025-07-09 15:42:15 Arisu /upload SUCCESS  
2025-10-20 12:25:44 Arisu /dashboard SUCCESS  
2025-02-19 10:50:15 Asimov /download SUCCESS  
2025-02-28 00:29:25 Asimov /dashboard SUCCESS  
2025-05-01 21:27:13 Asimov /login DENIED  
2025-08-25 00:54:42 Asimov /logout DENIED  
2025-03-10 15:47:13 Edward /login SUCCESS  
2025-03-16 00:28:44 Edward /404 DENIED  
2025-05-05 00:30:57 Edward /settings DENIED  
2025-06-22 14:50:54 Edward /404 FAIL  
2025-07-02 06:18:13 Edward /login DENIED  
2025-08-26 21:23:44 Edward /login FAIL  
2025-12-28 03:59:56 Edward /logout DENIED  
2025-03-13 05:12:58 Ein /404 SUCCESS  
2025-04-15 22:53:29 Ein /upload DENIED  
2025-04-23 23:51:43 Ein /upload SUCCESS  
2025-05-28 02:25:40 Ein /download DENIED  
2025-07-01 06:35:15 Ein /404 FAIL  
2025-09-18 03:35:36 Ein /upload SUCCESS  
2025-01-02 15:16:38 Eren /download FAIL  
2025-01-15 22:59:18 Eren /settings DENIED  
2025-04-26 10:19:38 Eren /download DENIED  
2025-08-02 00:23:04 Eren /admin FAIL  
2025-11-01 05:17:40 Eren /settings DENIED  
2025-01-19 11:35:44 Faye /upload DENIED  
2025-02-19 05:43:40 Faye /logout DENIED  
2025-03-04 10:47:58 Faye /dashboard DENIED  
2025-03-09 23:04:42 Faye /404 DENIED  
2025-07-19 22:59:38 Faye /reset FAIL  
2025-10-16 04:38:31 Faye /reset DENIED  
2025-12-13 12:02:17 Faye /logout DENIED  
2025-01-23 22:41:06 Haruka /dashboard FAIL  
2025-01-16 03:02:14 Haruka /reset SUCCESS
```

Q8

```
1: japariona@cis106: ~  
japariona@cis106:~$ sort -k4,4 -o sorted_access_log_by_endpoint.log access.log  
japariona@cis106:~$ cat sorted_access_log_by_endpoint.log  
2025-02-20 09:26:26 Julia /404 FAIL  
2025-03-09 23:04:42 Faye /404 DENIED  
2025-03-13 05:12:58 Ein /404 SUCCESS  
2025-03-16 06:28:44 Edward /404 DENIED  
2025-04-08 14:17:14 Kei /404 DENIED  
2025-05-02 01:46:24 Arisu /404 SUCCESS  
2025-05-05 14:31:33 Jet /404 FAIL  
2025-06-22 14:50:54 Edward /404 FAIL  
2025-07-01 06:35:15 Ein /404 FAIL  
2025-08-03 02:10:40 Rize /404 SUCCESS  
2025-09-06 10:17:57 Haruka /404 SUCCESS  
2025-09-09 13:40:49 Ken /404 DENIED  
2025-05-13 05:05:05 Annie /admin DENIED  
2025-06-16 05:20:44 Rei /admin DENIED  
2025-07-13 11:50:47 Haruka /admin SUCCESS  
2025-07-26 15:05:46 Seras /admin SUCCESS  
2025-08-02 08:23:04 Eren /admin FAIL  
2025-08-04 03:49:27 Shinji /admin SUCCESS  
2025-10-01 10:03:06 Touka /admin DENIED  
2025-11-07 12:59:18 Julia /admin DENIED  
2025-12-01 15:13:12 Jet /admin SUCCESS  
2025-12-25 21:14:04 Ken /admin SUCCESS  
2025-01-23 22:41:06 Haruka /dashboard FAIL  
2025-02-19 23:32:07 Ken /dashboard DENIED  
2025-02-28 00:29:25 Asimov /dashboard SUCCESS  
2025-03-04 10:47:58 Faye /dashboard DENIED  
2025-09-12 20:38:53 Spike /dashboard DENIED  
2025-10-20 12:25:44 Arisu /dashboard SUCCESS  
2025-11-20 09:30:50 Honami /dashboard DENIED  
2025-12-02 05:03:44 Shinji /dashboard FAIL  
2025-01-02 15:16:38 Eren /download FAIL  
2025-01-19 13:53:56 Arisu /download SUCCESS  
2025-01-19 21:04:13 Seras /download DENIED  
2025-02-19 19:50:15 Asimov /download SUCCESS  
2025-03-18 03:38:41 Kei /download DENIED  
2025-04-04 21:44:10 Annie /download FAIL  
2025-04-22 07:36:32 Ken /download SUCCESS  
2025-04-26 10:19:38 Eren /download DENIED  
2025-05-22 18:18:45 Honami /download FAIL  
2025-05-28 02:25:40 Ein /download DENIED  
2025-11-15 06:16:09 Rize /download FAIL  
2025-11-25 04:07:36 Touka /download DENIED
```

Q9

```
1: japariona@cis106: ~  
japariona@cis106:~$ cut -d' ' -f3,4 --output-delimiter=' accessed: ' access.log  
Eren accessed: /download  
Asimov accessed: /download  
Jet accessed: /login  
Spike accessed: /upload  
Rei accessed: /login  
Honami accessed: /download  
Arisu accessed: /404  
Edward accessed: /settings  
Touka accessed: /settings  
Shinji accessed: /reset  
Ken accessed: /admin  
Haruka accessed: /admin  
Ken accessed: /upload  
Jet accessed: /404  
Rize accessed: /logout  
Julia accessed: /settings  
Rei accessed: /admin  
Honami accessed: /reset  
Annie accessed: /download  
Seras accessed: /admin  
Edward accessed: /logout  
Annie accessed: /upload  
Faye accessed: /reset  
Haruka accessed: /reset  
Spike accessed: /dashboard  
Rize accessed: /upload  
Rize accessed: /404  
Annie accessed: /upload  
Faye accessed: /reset  
Eren accessed: /admin  
Touka accessed: /logout  
Ken accessed: /404  
Ein accessed: /upload  
Rize accessed: /logout  
Shinji accessed: /admin  
Julia accessed: /404  
Ken accessed: /logout  
Ein accessed: /404  
Faye accessed: /dashboard  
Shinji accessed: /logout  
Arisu accessed: /upload  
Annie accessed: /settings  
Eren accessed: /upload
```